Final Project	
Github Portfolio	
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Course Code and Title: CPE201A - Computer System Administration and Troubleshooting	**Instructor: Sir NJ Lusung**

1. Objective/s:

This final project aims to demonstrate the student's ability to create and build a GitHub portfolio by compiling and organizing outputs from other Computer Engineering courses.

2. Intended Leaning Outcome/s:

By the end of this final project, the students should be able to:

- Develop a professional GitHub portfolio that effectively showcases their projects and demonstrates integration of knowledge from various Computer Engineering courses.

3. Directions:

1. **Create a New Repository**
   - Log in to your GitHub account.
   - Click New Repository.
   - Set the repository title as:
     CPE201A_FP_SURNAME
   - Add a short description, for example: "Final Projects Compilation for CpE Courses."
   - Choose the repository visibility (Public or Private).
   - Click Create Repository.

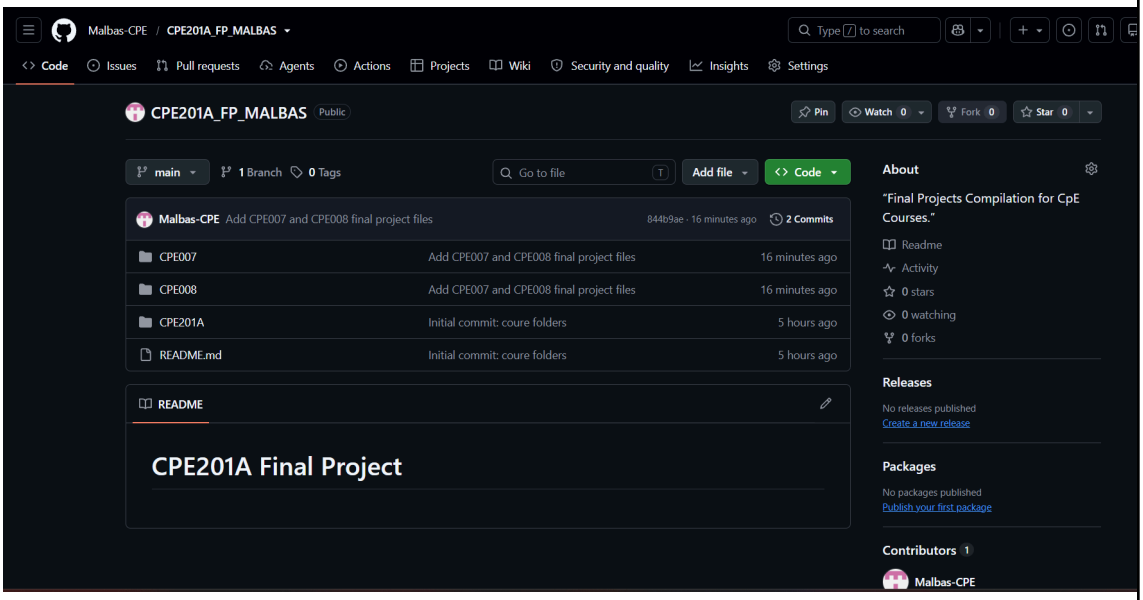
2. **Prepare Files on Ubuntu Linux**
   - Open your Ubuntu terminal.
   - Navigate to the directory where your final projects are saved.
     cd ~/Documents/CpE_Projects
   - Organize your project folders according to different CpE courses, for example:

```
CPE201A_FP_SURNAME/
├── CPE101/
├── CPE102/
├── CPE103/
└── CPE104/
```

   - Make sure each folder contains the corresponding final project files from that course.

3. **Initialize Git and Push to GitHub**
4. **Verify and Share**
    - Go to your GitHub repository online and confirm that all files and folders are properly uploaded and organized.
    - Copy your repository link and make sure it is accessible (if required, set repository visibility to “Public”).
5. **Documentation**
    - Take screenshots of the following:
        - Repository creation page.
        - Organized folder structure.
        - Ubuntu terminal commands during initialization, commit, and push.
        - Final GitHub repository page showing the uploaded projects.
    - Compile all screenshots and input it in Section 4. Outputs.

4. Outputs:

1. 

2.

```
malbas-cpe201a@malbas-cpe201a-VirtualBox:~$ cd
malbas-cpe201a@malbas-cpe201a-VirtualBox:~$ mkdir CPE201A_FP_MALBAS
malbas-cpe201a@malbas-cpe201a-VirtualBox:~$ cd CPE201A_FP_MALBAS
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ mkdir CPE007 CPE008 CPE201A
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls
CPE007  CPE008  CPE201A
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls-R
ls-R: command not found
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls -R
.:
CPE007  CPE008  CPE201A

./CPE007:

./CPE008:

./CPE201A:
```

3.

```
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ sudo apt install git -y
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
git is already the newest version (1:2.53.0-1ubuntu1.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 95
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git config --global user.name "Malbas-CPE"
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git config --global user.email "qsgmalbas26@tip.ec
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ echo "# CPE201A Final Project" > README.md
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ touch CPE007/.gitkeep CPE008/.gitkeep CPE201A/.git
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls -a
.  ..  CPE007  CPE008  CPE201A  README.md
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/malbas-cpe201a/CPE201A_FP_MALBAS/.git/
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git branch -M main
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git add .
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git status
```

```
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ read -s TOKEn github_pat_11CRPOYQA0biHhJBiWNVln_CoJUI7oDnnxxT9yZhq32SbVHGIvjhN
jsLQ4I3LDORfFLZNarC
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ echo ${#TOKEN}
93
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ^C
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git remote set-url origin https://Malbas-CPE:${TOKEN}@github.com/Malbas-CPE/CP
ALBAS.git
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git remote set-url origin https://Malbas-CPE:${TOKEN}@github.com/Malbas-CPE/CP
ALBAS.git
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git push -u origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Malbas-CPE/CPE201A_FP_MALBAS.git/'
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git push -u origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Malbas-CPE/CPE201A_FP_MALBAS.git/'
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ssh-keygen -t ed25519 -C "qsgmalbas26@tip.edu.ph"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/malbas-cpe201a/.ssh/id_ed25519):
Enter passphrase for "/home/malbas-cpe201a/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/malbas-cpe201a/.ssh/id_ed25519
Your public key has been saved in /home/malbas-cpe201a/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:tKJ1lsMS0bUOiBdE+jOKim8ZsH9Pzx6FlpJbhN+ckZ0 qsgmalbas26@tip.edu.ph
The key's randomart image is:
+--[ED25519 256]--+
|     o=. ..      |
|     o =.  + .   |
|    o = + + E    |
|.    o B X o     |
| o    X S *      |
|. .. + % o       |
| ..oo o .        |
|..+ .. o .       |
|oo.. ...+        |
+----[SHA256]-----+
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINGScCJeSKcJBfIMva2YugIta6zFC7aDtxLKMQB8PxyQ qsgmalbas26@tip.edu.ph
```

```
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git remote set-url origin git@github.com:Malbas-CPE/CPE201A_FP_MA
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git push -u origin main
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINGScCJeSKcJBfIMva2YugIta6zFC7aDtxLKMQB8PxyQ qsgmalbas26@tip.edu.ph
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ^C
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ^C
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ^C
bash: :s^C: substitution failed
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ sudo apt install xclip -y
[sudo: authenticate] Password:

sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Installing:
  xclip

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 95
  Download size: 17.2 kB
  Space needed: 54.3 kB / 67.1 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute/main amd64 xclip amd64 0.13-4build1 [17.2 kB]
Fetched 17.2 kB in 0s (45.0 kB/s)
Selecting previously unselected package xclip.
(Reading database… 152046 files and directories currently installed.)
Preparing to unpack …/xclip_0.13-4build1_amd64.deb…
Unpacking xclip (0.13-4build1)…
Setting up xclip (0.13-4build1)…
```

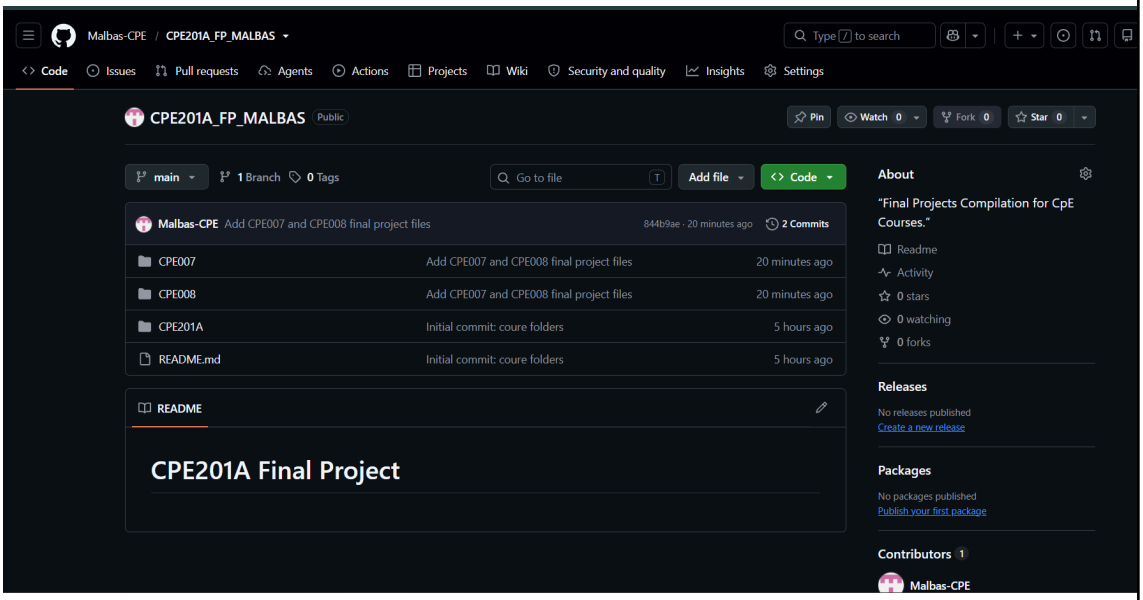
```
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cd ~/CPE201A_FP_MALBAS
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls ~/Downloads
'GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp'  'Interview Group 2.pdf'
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cp ~/Downloads/"GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp"
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cp ~/Downloads/"Interview Group 2.pdf" CPE008/
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls CPE007 CPE008
CPE007:
'GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp'

CPE008:
'Interview Group 2.pdf'
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git add .
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git status
On branch main
Your branch is up to date with 'origin/main'.

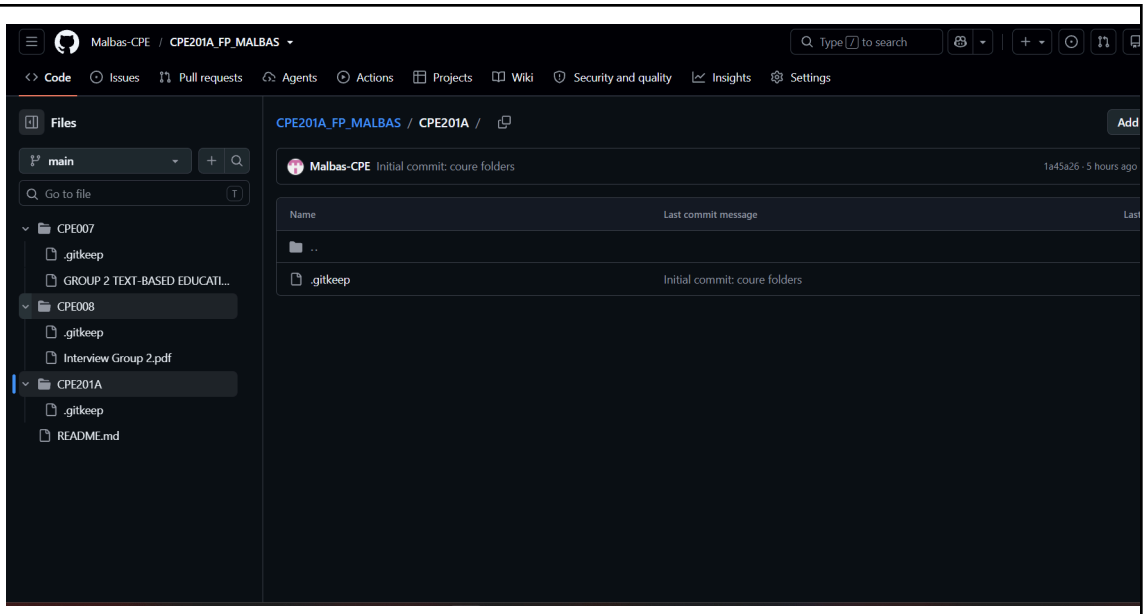
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   CPE007/GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp
        new file:   CPE008/Interview Group 2.pdf

malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git commit -m "Add CPE007 and CPE008 final project files
[main 844b9ae] Add CPE007 and CPE008 final project files
 2 files changed, 3153 insertions(+)
 create mode 100644 CPE007/GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp
 create mode 100644 CPE008/Interview Group 2.pdf
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git push
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 116.45 KiB | 424.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Malbas-CPE/CPE201A_FP_MALBAS.git
   1a45a26..844b9ae  main -> main
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$
```

4.



5.




```
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cd ~/CPE201A_FP_MALBAS
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls ~/Downloads
'GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp'  'Interview Group 2.pdf'
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cp ~/Downloads/"GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp"
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ cp ~/Downloads/"Interview Group 2.pdf" CPE008/
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ ls CPE007 CPE008
CPE007:
'GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp'

CPE008:
'Interview Group 2.pdf'
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git add .
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   CPE007/GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp
        new file:   CPE008/Interview Group 2.pdf

malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git commit -m "Add CPE007 and CPE008 final project files
[main 844b9ae] Add CPE007 and CPE008 final project files
 2 files changed, 3153 insertions(+)
 create mode 100644 CPE007/GROUP 2 TEXT-BASED EDUCATIONAL GAME.cpp
 create mode 100644 CPE008/Interview Group 2.pdf
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$ git push
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 116.45 KiB | 424.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Malbas-CPE/CPE201A_FP_MALBAS.git
   1a45a26..844b9ae  main -> main
malbas-cpe201a@malbas-cpe201a-VirtualBox:~/CPE201A_FP_MALBAS$
```

5. Conclusion/Learnings/Analysis:

I learned the different terminals in Ubuntu and how to use them.

6. Assessment Rubric:

Rubric for SO 7 (7)

Criteria	Ratings						Pts
SO 7 PI 1 **Student Outcome 7.1** Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 3 **Student Outcome 7.3** Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts
SO 7 PI 4 **Student Outcome 7.4** Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6 pts

Total Points: 18